

Does a frozen galaxy encoder know that turning a picture changes nothing?

We took the **Symmetry** question and went deep rather than broad. Two of our expectations were falsified by controls, and one of our own headline numbers turned out to be half measurement error. A **second track** asks the complementary question with an energy-based model: not how the representation moves, but what it encodes and how tightly.

team: claude maxing · Jack O'Brien · Murman Grugenidze · AION-1-Large, frozen, image-only, 1024-d

Rotation changes nothing physical about a galaxy. Does the embedding agree?

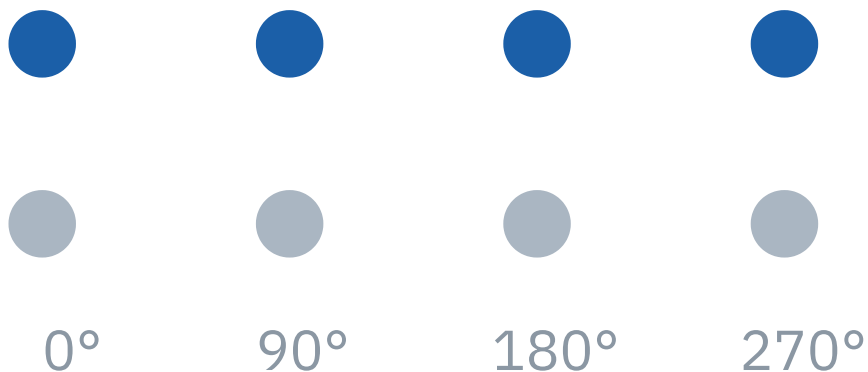
- **Re-encoded 512 real galaxies at 16 position angles, and mirrored** — 9,216 passes through the frozen encoder, so each galaxy's rotation orbit is *measured*, not assumed.
- **Nothing constrained the answer in advance.** The AION-1 paper never mentions rotation, flip or augmentation — full invariance and full equivariance were both live.
- **Two confounds had to die first.** Arbitrary rotation resamples pixels; and 90° is the square pixel grid's *own* symmetry, so an apparent 4-fold signal is exactly what an artefact looks like.
- **Then: is reflection just rotation?** A mirror reverses a spiral's winding and changes nothing else — it isolates chirality.
- Scope, honestly: we answered the mirror-vs-resampling half. We did **not** reach the cross-domain half.



Controls that cost nothing, and a substrate that records every choice

MEASUREMENT

- **Angular harmonic decomposition** separates the rotation-invariant part of the orbit from what actually moves.
- **Exact transforms are free controls.** The 16-angle grid already contains one interpolation-free sub-orbit and three resampled ones *at identical geometry* — so this control cost **zero GPU time**.
- **A null, not a threshold.** Interleaving the angles yields points *known* to be on the orbit, measured the same way — which is what turns "the mirror is close" into a chirality test.
- **TwoNN dimension, corrected for sample size** by subsampling the full cloud to the quotient's own n , 64 times.



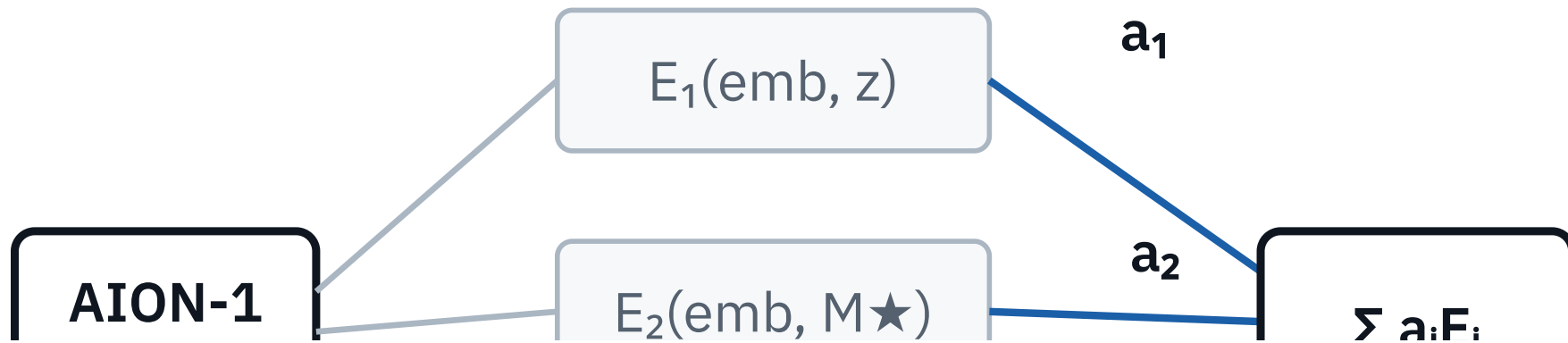
phase 0 — no resampling

phases 1–3 — resampled

Not how the representation moves, but what it encodes – and how tightly

- **One energy per property, summed:** $E = \sum a_i E_i(\text{emb}, x_i)$. Each E_i is an independent network; the a_i are learned and act as Lagrange multipliers on the components.
- **Generative, not discriminative.** A probe regresses one target at a time. This models embedding and parameters *jointly*, so properties compete for the same representation.
- **Posteriors, not point estimates.** Nested sampling over the energy returns a full distribution per parameter – "how well is this pinned down" is answered by a *width*, not an R^2 .
- The a_i are an **interpretability readout**, and the per-parameter input projections show *which* embedding directions carry each property.
- 8 catalogue parameters: redshift, `logMstar`, `logMform`, `avgSfr`, `tageMw`, `zMw` and errors.

THE MODEL



Two expectations falsified, one number halved

Largely rotation-invariant — untrained

Emergent, not designed. Of what varies, $m=2$ (180°) dominates — a disc ellipse or bar.

80.9%

POWER INVARIANT

That signal is real, not resampling **FALSIFIED**

Interpolation-free vs resampled sub-orbits. Resampling accounts for 0.45%.

0.996

EXACT ÷ RESAMPLED

Chirality is not encoded **FALSIFIED**

The mirror lands on the galaxy's own orbit. No chiral tail: 45.7% — a coin flip.

0.973

MIRROR ÷ NULL

Orientation costs ~2.4 of ~14 dimensions **SELF-CORRECTED**

Not 4.32 — the clouds were measured at $n=8,058$ and $n=512$; 1.94 was estimator bias.

2.37

± 0.60 DIMS

Orientation-free properties do *not* stay still

Every readout wobbles 0.20–0.42 of its own error. Bulge dominance is the lone exception at 0.002.

0.46

DEX, MASS, P-TO-P

Limits. 16 angles resolves only to $m=8$ (Nyquist) and $m=4$ already carries 17.9%. 512 galaxies, not 10,000 — GPU *time*, not memory; that sample size is what forced the correction above. **gzArm** ships no images, so chirality is testable only geometrically. Cross-domain half unrun: the GPU throttled to 270 MHz at 88 °C.

37.7%

